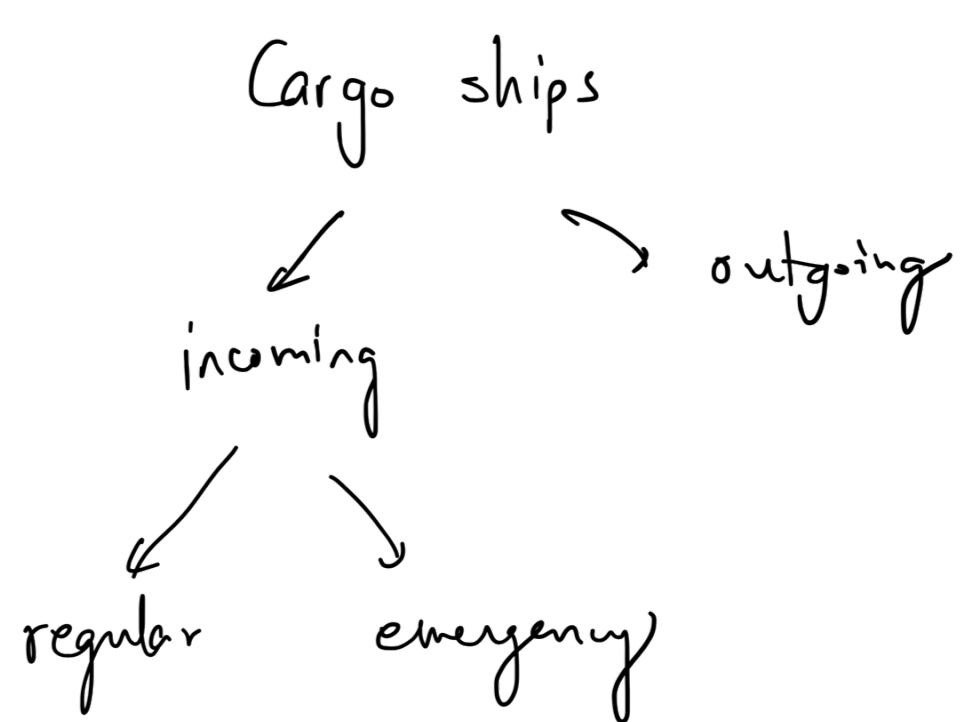




→ Before undocking, all ships have to guess a secure radio channel frequency to connect to Jack & tell him that the ship is leaving.

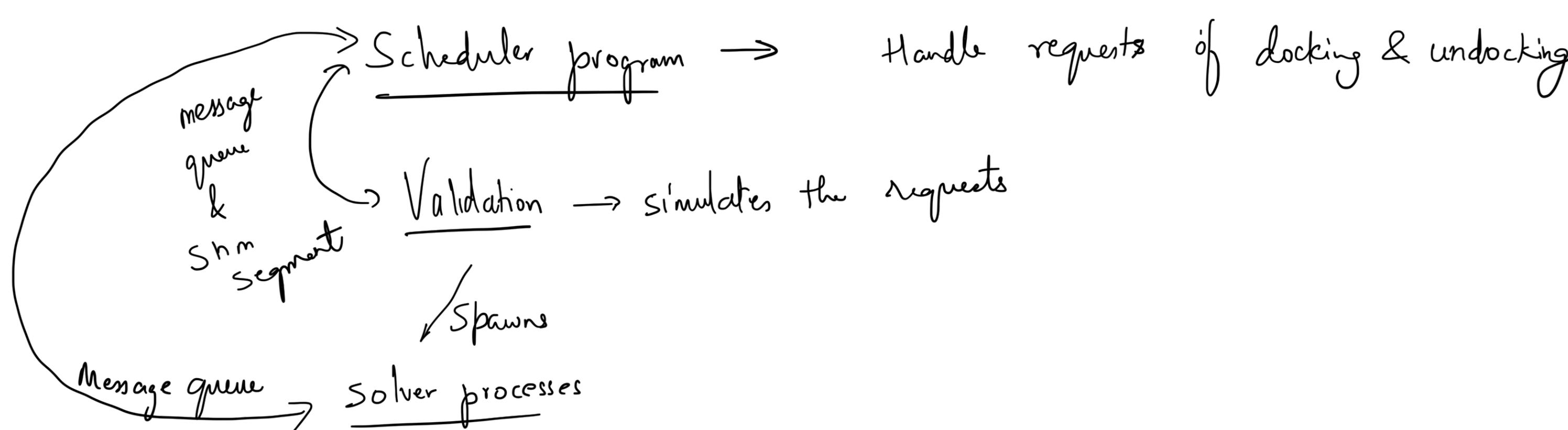
Incoming ship:- docked → unloaded → undocked

Outgoing:- docked → loaded → undocked

Dock of category X → has X cranes on it.

Ship category $\leq$ Dock's category

Each ship's cargo → array of boxes of varying weights.



Ship Request

- Ship ID → Waiting Time
- Thrustup → num Cargo
- Category → Cargo (max Cargo count)
- Direction
- Emergency

Ship ID + Direction → unique identification of ship.

only consider for incoming ships

0 1 2 2
2 3 2 2
2 2 3 3

t₁ → Ship (01) and ship (0-1)

t₂ → dock both ships

Ship 01 → assigned 2² [0, 3, 0, 3, 0, 3, 0, 0, 0, 4, 0, 4]
Ship 0-1 → assigned 2³ [1, 4, 1, 2, 1, 2, 1, 3, 1, 3, 1, 2, 1, 4]

2, 4 [10 | 4] → [7 | 7]
t₂ → t₉ occ → t₀ → free

1 4
9 | 5
7 | 7

t₂ → t₉ occ.
t₁₀ → free

t₂ → ship (1-1) cat 1 → 3⁰
ship (2-1) cat 1 → 3¹

Ship (1-1) → [0, 4, 0, 0, 0, 2, 0, 4, 0, 0, 0, 2, 0, 3]

2 4 1
3 | 2 | 9
4 | 5 | 5 → max → 5

t₂ → t₇ (this will be occ) → t₈ → free

Ship (2-1) → [0, 3, 0, 4, 0, 2, 0, 3, 0, 2, 0, 0]

4 4 1
2 | 3 | 7
4 | 4 | 4 → max 4 → t₂ → t₆ occ → t₇ → free

QUEUE

- ~~(t₁) ship (2-1) → WT = 0 → Emergency~~
- ~~(t₂) ship (3-1) → WT = ∞~~
- ~~(t₄) ship (1-1) → WT = 6~~
- ~~(t₅) ship (3-1) → WT = 7~~
- ~~(t₆) ship (4-1) → WT = ∞~~
- (t₇) ship (4-1) → WT = 6

at t₇ → dock 3¹ gets free → allot ship (2-1)

(t₈) → ship (6-1) → WT = 7

at t₈ 3⁰ gets free → ship (3-1)

[1, 4, 0, 3, 0, 3, 0, 4, 0, 4, 0, 2]

4 4 1
2 | 3 | 7
4 | 4 | 4

t₇ → t₁₁ → t₁₂ free

(t₉) → no ship requests

(t₁₀) → no ship requests, but 2² and 2³ get free

Ship (1-1) → get allotted dock 2² → [0, 3, 0, 3, 0, 2, 0, 0, 0, 3, 0, 2, 0, 3]

Ship (3-1) → cannot be allotted to 2³ as category is 3, so has to wait for dock with cat = 3

1 4
8 | 6
7 | 7

t₁₀ → t₁₇ → t₁₈ (free)

Ship (4-1) → allot to 2³

[1, 1, 1, 2, 1, 2, 1, 0, 1, 0, 1, 0, 1, 0]

2 4
10 | 4
7 | 7

t₁₁ → no ship, everything busy

t₁₂ → No ship, dock 3¹ gets free

allot to ship (3-1)

[0, 3, 0, 0, 0, 2, 3, 0, 2, 0, 0, 0, 4, 0, 2, 4, 0, 2, 2, 0]

4 4 1
4 | 5 | 12
7 | 7 | 7

t₁₂ → t₁₉ → t₂₀

t₁₃ → No ship, busy

t₁₄ → No ship, busy, ship (4-1) leaves

t₁₅ → ship (4-1) leaves → WT = 6, dock 3⁰ is free ⇒ allot to ship (4-1)

t₁₅ → ship (5-1) arrives → WT = 6

Ship (4-1) → [0, 0, 0, 4, 0, 3, 0, 4, 0, 4]

2 4 1
1 | 4 | 5
2 | 4 | 1

t₁₅ → t₁₉ → t₂₀

t₁₆ → No ship requests, busy

~~t₁₇ → ship (6-1) arrives → WT = 7~~

t₁₈ → No ship requests → docks 2² and 2³ get free

Ship (5-1) → allot to 2² → [0, 2, 0, 3, 0, 3, 0, 2, 0, 0, 0, 2]

t₁₈ → t₂₄ → t₂₅

Ship (6-1) → allot to 2³

[0, 2, 0, 2, 0, 1, 0, 0, 2, 0, 0, 0, 4, 0, 3]

2 4
2 | 4
7 | 7

t₁₈ → t₂₅ → t₂₆